

Legal Revisions Regarding Chemical Substance Management in Japan and Associated Changes in Exposure Assessment Methods: Current Status and Future Challenges

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Outline of Presentation

Legal Revisions Regarding Chemical Substance Management in Japan and Associated Changes in Exposure Assessment Methods : Current Status and Future Challenges

1. Legal Revisions Regarding Chemical Substance
2. Workplace Monitoring/measurement Methods
3. Future Challenges and Summary

Note

Occupational Health Regulations in Japan

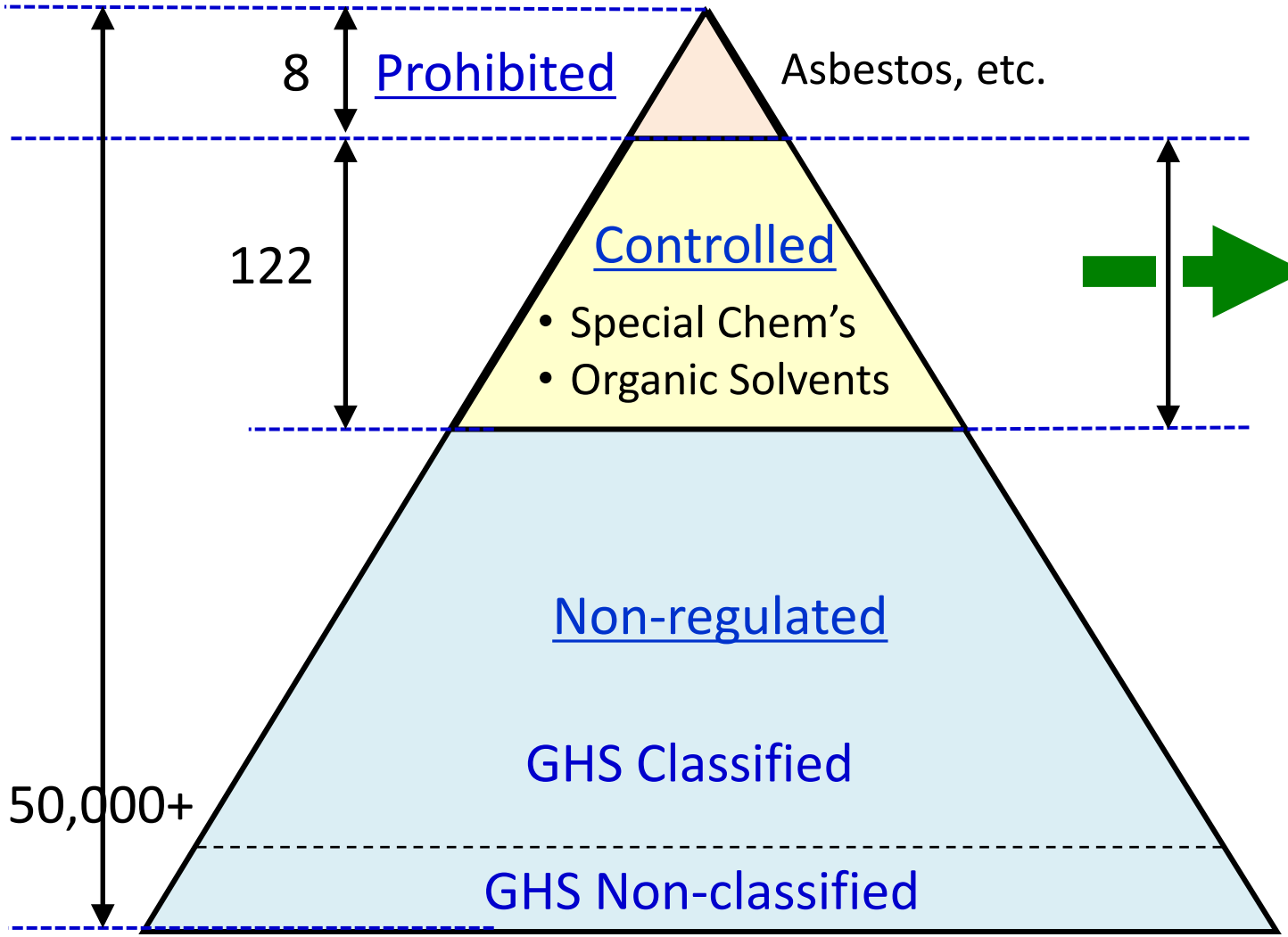
- Long history of OH regulations（100+ yrs)
- Industrial Safety & Health Law (1972)
  - Forcing detailed rules, complex, demanding
- Limited # of chemicals specifically focused

Highly Regulated	Non (or, weakly) -regulated
Prohibited (8)	Other chemicals
Controlled (122) • Special chemical substances (~70) • Organic solvents (~ 50) • Lead, dust	

- Employers' mindset
  - Aiming to fulfil the minimum regulatory requirements
  - Weak self-motivation for protecting workers

Note

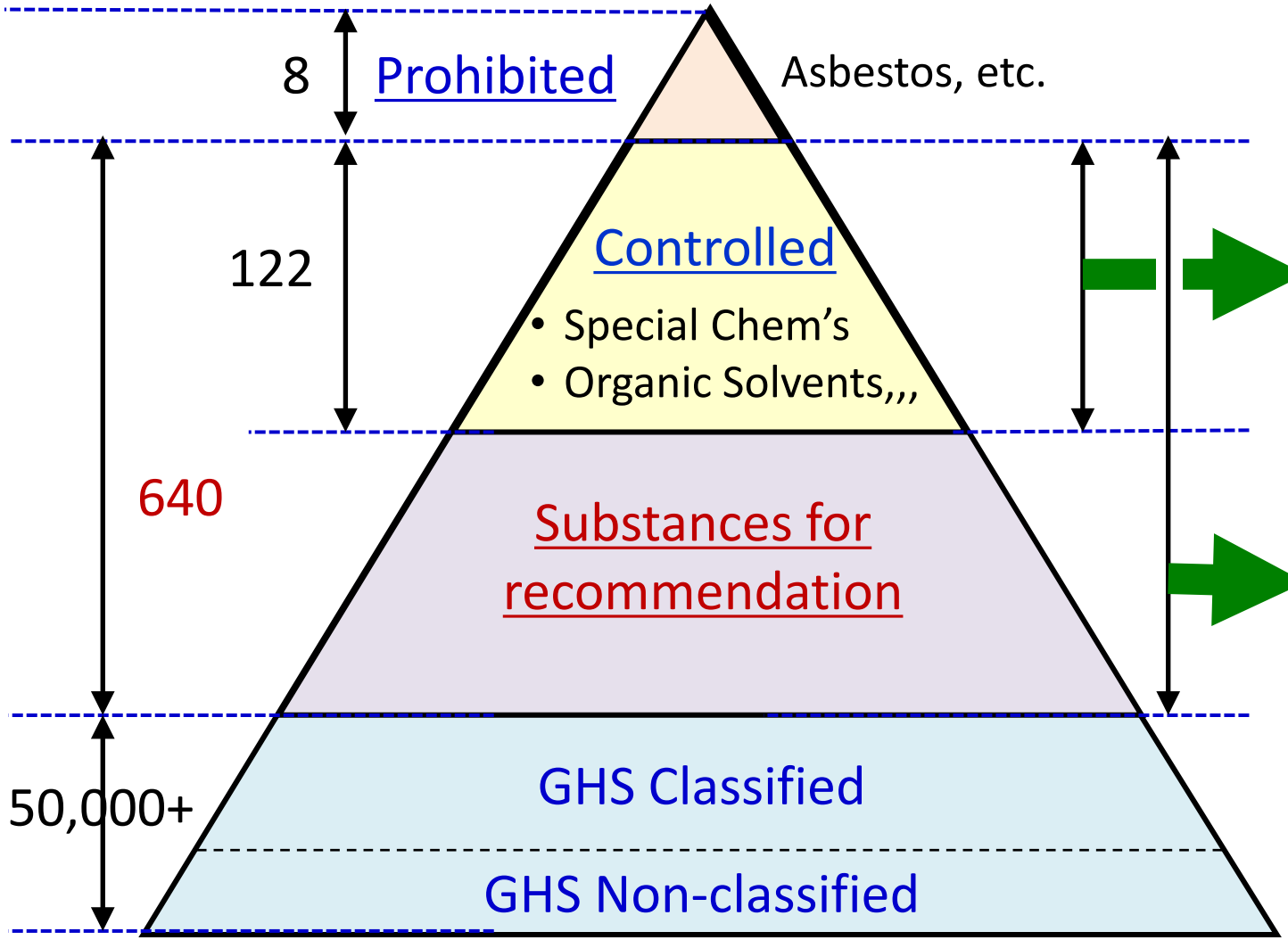
Structure of Chem. Management (before 2005)



- Detailed requirements
  - Environmental monitoring, twice/y
  - LEV for every vapor sources
  - Health surveillance, twice/y
    - For all workers handling chemicals

- Uniform/blanket approach, regardless of actual risks
  - Not rational/scientific

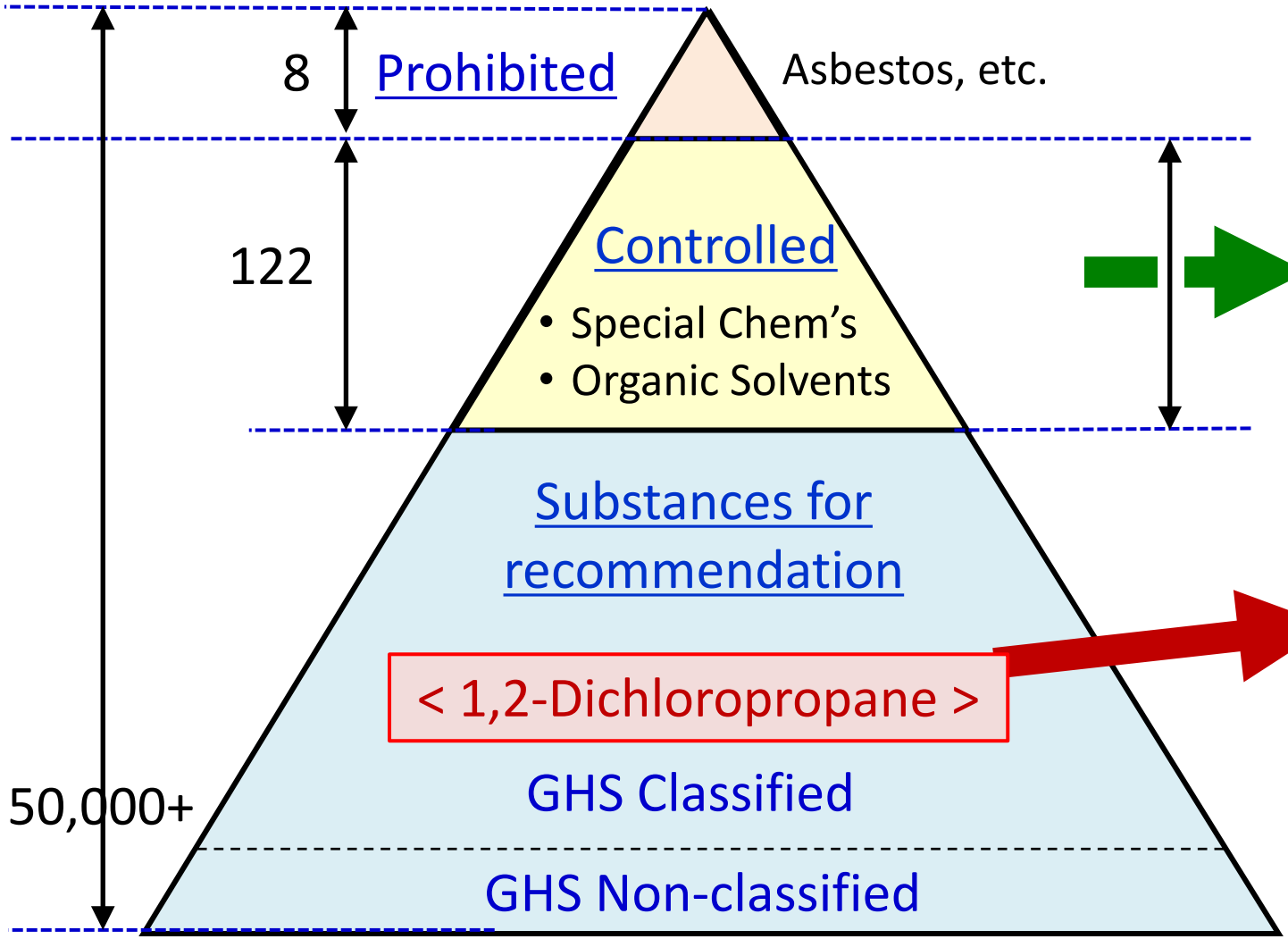
Structure of Chem. Management (2006-2015)



- **Detailed requirements**
  - Environmental monitoring, twice/y
  - LEV for every vapor sources
  - Health surveillance, twice/y
    - For all workers handling chemicals

- **Recommended (after 2006)**
  - Risk assessment
    - Not widely practiced
  - GHS labeling, SDS provision

“Bile Duct Cancer” Incident in 2012



- ◆ Requirements - Detailed
  - – Environmental monitoring, twice/y
  - – LEV for every vapor sources
  - – Health surveillance, twice/y
    - • All workers who handle the chemical etc.

- ◆ Printing factories
  - – Used 1,2-DCP widely
  - – To avoid regulatory duties

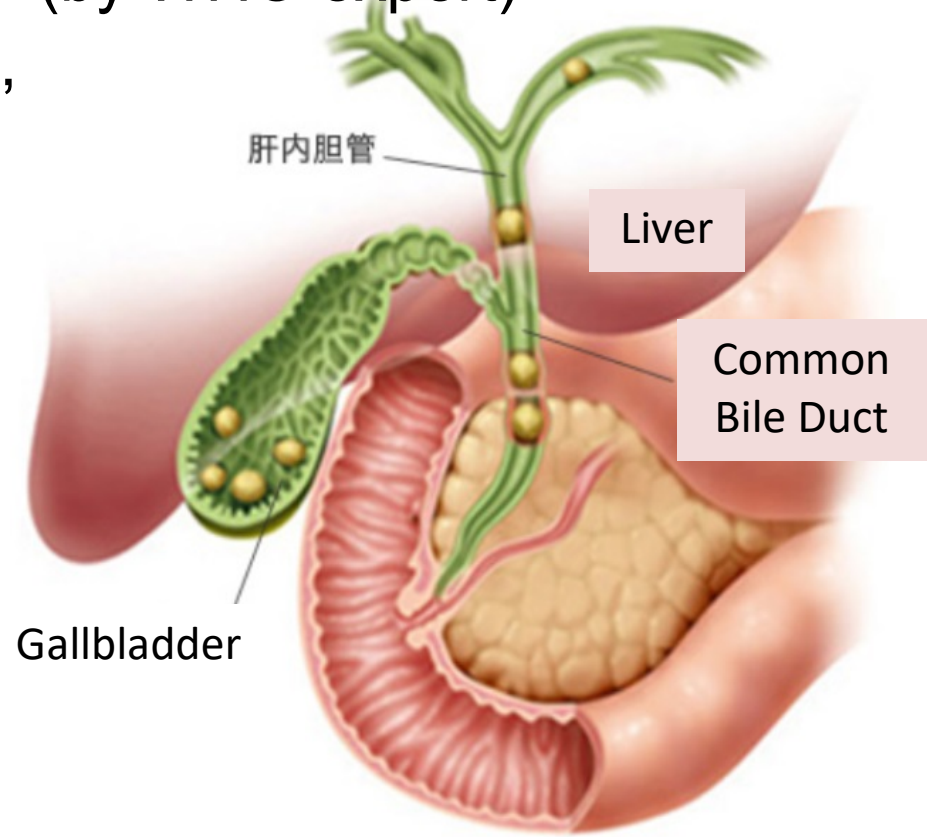
Cancer incident

“Bile Duct Cancer” Incidents

- In 2012～2014
- Workers (50+) in printing industry died
- Major cause：1,2- Dichloropropane (a cleaning solvent)
  - “Why such an incident had occurred in Japan !?” (by WHO expert)
  - The employers did not provide exposure control,
    - Because 1,2-DCP was NOT regulated.



Work in printing industry (Image)



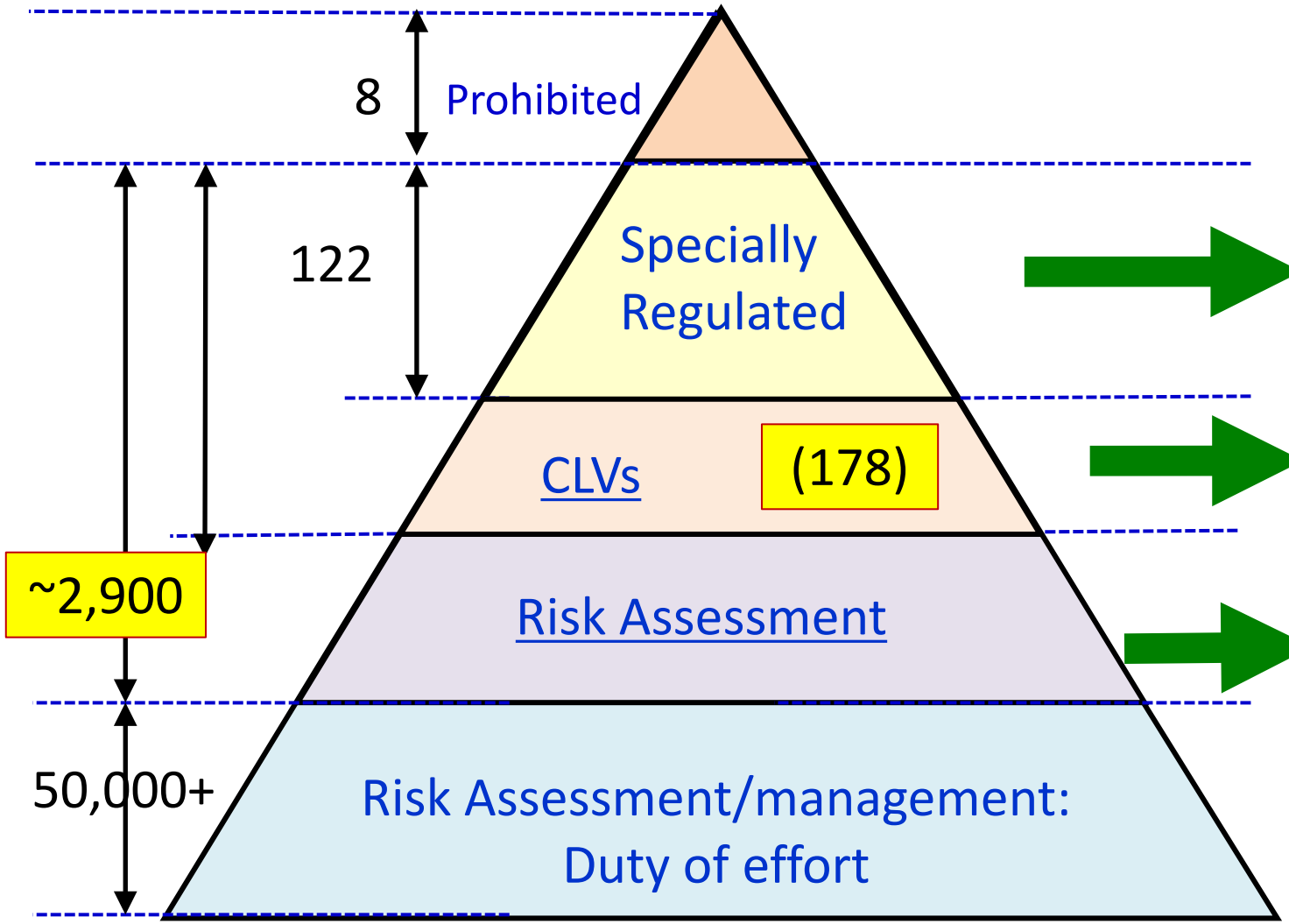
Other Cancer Cases: Caused by Non-regulated chemicals

Year	Cause	Manufacturing work	Exp. Route	Types of cancer
2015	o-Toluidine	Intermediates for dyes/pigments	Skin	Bladder
2016~	MOCA : 4,4'-Methylenebis(2-chloroaniline)	Chemicals using aromatic amines	Skin	Bladder

◆ Leading to strengthening regulations regarding chemical-protective gloves, after 2022.

Structure of Chem. Management: 2016~ / 2024 Fully-established

(# of chemicals, 2026)



- ◆ Continuing
- ◆ Having government-defined CLVs
  - To be ~900 in several years

CLVs: Concentration Limit Values

- ◆ Risk assessment/ management mandatory
- ◆ GHS labeling, SDS provision

“New Chemical Management” by the Government: 2024~

- The government’s policy report (July, 2021)
  - Regulatory system to be changed drastically
    - Conventional policy: “Forcing detailed rules”
      - » Not sufficient for preventing workplace incidents



    - New: “Autonomous management by employers”
      - » Employers bearing full responsibility for accident/ illness



  - Full regulations established/ implemented (Oct, 2024).

Note

“New Chemical Management”: Employers’ Action (mandatory)

1. Communication of chemical hazards to workers
   - GHS labeling, SDS provision
   - For designated chemicals: ~ 1,900
   - Worker training

2. Risk assessment and Management
   - For designated chemicals: ~ 1,900
   - Methods to be determined by employers ← “Autonomous management“
   - Minimize exposure as much as possible



   - Exposure ≦ OELs, for chemicals with OELs
     - Exposure assessment/management methods; at the employer's discretion.

Occupational Exposure Limit (OEL)

“New Chemical Management”: Employers’ Action (mandatory)

3. Compliance for substances with “Concentration Limit Values (CLVs *)"
   - (*) CLVs: Set by the government
   - 178 substances (Oct/2026)
   - Exposure ≦ CLVs,.
     - Applies only to indoor workplaces.

4. Health surveillance
   - To be conducted only when necessary, at the employer’s discretion
     - When workers facing a high risk, based on risk assessment

Concentration Limit Values (CLVs)(OEL)

“New Chemical Management”: Employers’ Action (mandatory)

5. Appointment of “Field supervisors” for Chemical Management

A) "Chemical Substance Supervisor"

– For all workplaces that handling chemicals

B) “PPE management Supervisor"

– For all workplaces where PPEs are used

Personal Protective Equipment (PPE)

6. Appointment of “Chemical Management Experts”

◆ Supporting/advising employers, especially for SMEs

A) “Chemical Substance Management Expert”

- Occupational Hygienists (JAWE, IOHA)
- Occupational Hygiene Consultants (Government qualification)

B) “Work Environment Management Expert”

“New Chemical Management”: Government Guidance etc.

1. Development of a tool for simplified risk assessment
   - ◆ CREATE-SIMPLE
     - – Highly user-friendly, Offering relatively accurate assessments
   - ◆ Widely utilized

STEP 4 リスクの判定　Risk Assessment Result

リスクを判定　実施レポートに出力

	Exposure Limit			Exposure, calculated			Risk Levels (I to IV)	
	吸入（8時間）	吸入（短時間）	経皮吸収	吸入（8時間）	吸入（短時間）	経皮吸収	吸入（8時間）	吸入（短時間）
1	67-64-1　アセトン　Acetone						有機則（第２種有機溶剤）、リスクレベルS	
	200 ppm	500 ppm	3561 mg/day	5～50 ppm	200 ppm	3.71 mg/day	Ⅱ-A	Ⅱ
2	78-93-3　メチルエチルケトン　MEK						有機則（第２種有機溶剤）、皮膚等障害化	
	75 ppm	150 ppm	-	5～50 ppm	200 ppm	-	Ⅱ-B	Ⅲ

“New Chemical Management”: Government Guidance

2. Prevention of dermal exposure

- Development of a selection guide for chemical-protective gloves

Glove material (Nitrile, Latex, Butyl etc.)/ Thickness

Permeation resistance

Chemicals (~ 1,300 cpds.)

耐透過性一覧

		材料	ニトリル	ニトリル	ニトリル	ニトリル	天然ゴム（ラテック	ブチル	クロロプレン	ポリビニルアルコール（PVA）
62-75-9	N，N－ジメチルニトロソアミン		×	×	×	×	×	○	×	×
63-25-2	N－メチルカルバミン酸１－ナフチル【カルバリル】		◎	◎	◎	◎	◎	◎	◎	◎
63-92-3	フェノキシベンザミン塩酸塩		◎	◎	◎	◎	◎	◎	◎	◎
64-04-0	ベーターフェニルエチルアミン（別名フェネチルアミン）		×	×	△	△	×	◎	×	◎
64-18-6	ぎ酸		×	△	△	△	△	◎	◎	×
64-19-7	酢酸		×	×	△*	△*	×	◎	○	×
64-67-5	硫酸ジエチル		×	△	○	○	×	◎	○	-
64-69-7	ヨード酢酸		◎	◎	◎	◎	○	◎	◎	◎
65-85-0	安息香酸		◎	◎	◎	◎	◎	◎	◎	◎
67 56 1	メタノール		×	×	△	△	×	◎	△	×
67-68-5	ジメチルスルホキシド		×*	△	○	○*	△	◎	△	×
67-72-1	ヘキサクロロエタン		◎	◎	◎	◎	◎	◎	◎	◎
68-11-1	メルカプト酢酸		×	×	△	△*	△	◎	△	×
68-12-2	N，N－ジメチルホルムアミド		×	×	△	△	×	◎	×	△
69-72-7	サリチル酸		◎	◎	◎	◎	◎	◎	◎	◎
70-25-7	N－メチル－N'－ニトロ－N－ニトロソグアニジン		◎	◎	◎	◎	◎	◎	◎	◎

“New Chemical Management”: Government Guidance

3. Fit testing of RPEs

- Mandatory for specified use (e.g. welding work)
  - Annually



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Exposure/ Environment Measurement Methods (2026)

- No.1: Conventional measurement
- No. 2 to 6: Started 2021 or later

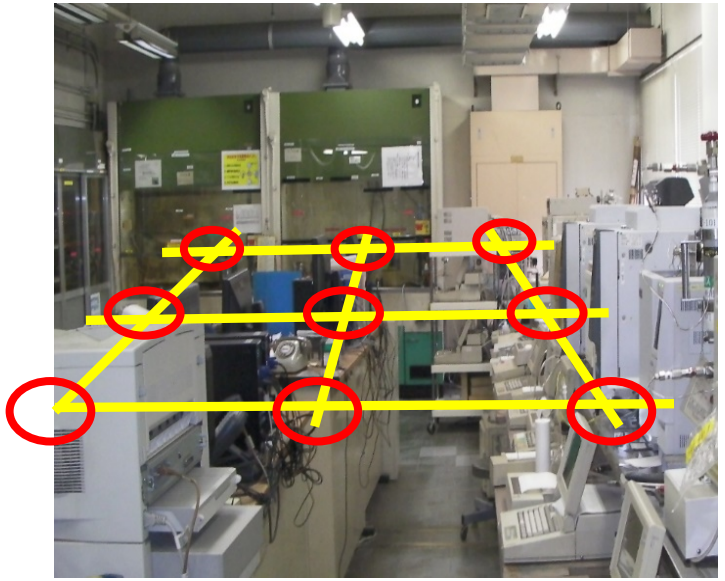
No.	Name of measurement	# of Chems	Scope of chemicals	Breasing Zone
1	A, B measurement	~90	Controlled Chemicals	–
2	C, D measurement	~90		○
3	Welding fume	1		○
4	Control-Class 3	~90		○
5	Confirmation measurement	178 *	Risk assessment mandatory	○
6	General exposure measurement	~800		○

* For chemicals with Concentration Limit Values (CLVs)

No. 1: Conventional “Work Environment (A, B-) Measurement (WEM) ”

- Area monitoring, regulatory required (1975~)
- Basic method (A-measurement)
  - Measuring grid points
  - # of samples: 5+
  - Measurement duration: 10+ minutes
  - Reference value: “Admin. Conc. Level”
  - “Control Class” judged by statistical analysis
    - Class 1: well-controlled
    - Class 2: requiring improvement
    - Class 3: requiring immediate action
- B-measurement: peak concentration

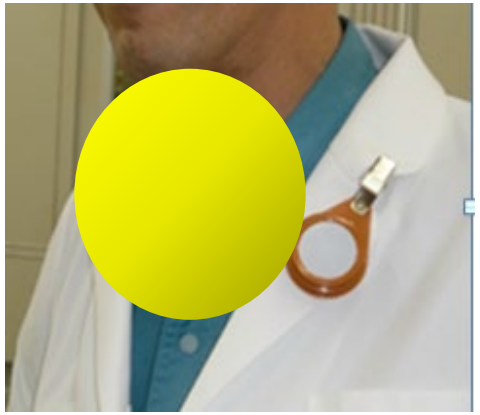
Work Environment Measurement (Image)



- Only “technicians” needed
  - Work Environment Measurement supervisor

No.2: C, D-Measurement

- C measurement
  - Using personal samplers
  - # of samples: 5+
  - Selecting workers from a "similar exposure work"
  - Measurement duration: 2+ hrs
  - D measurement: peak exposure
  - “Control Class” judged, same as A,B-measurement



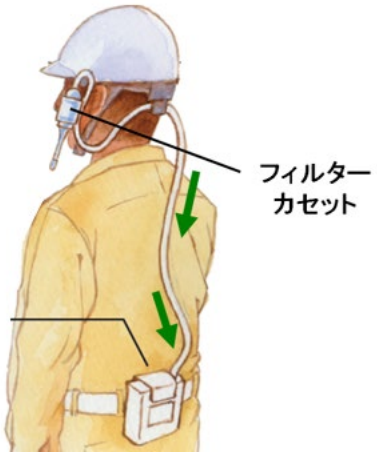
- Not a “personal exposure monitoring”
  - No 8-hour averaging
  - “Static monitoring” at the breathing zone

Note: Having a poor reputation and rarely used.

No. 3: Welding Fume Monitoring (from 2021)

- Personal sampling mandatory for welding only
  - High concentration in the breathing zone: close to the fume source
- Target Chemical: Manganese
  - OEL: 0.05mg/m3 as Mn
- # of samples: 2+
- Sampling duration: during the task
- Monitoring frequency: at least once in a workplace
- Result evaluation
  - Non-8hr-averaging
  - Comparing the highest value vs. OEL





Note: Monitored extensively in 2021–23, but is now rarely performed.

No. 4: “Control-Class 3” Measurement

- Control Class 3: requiring immediate corrective action
  - Result of a conventional WEM

↓

- If any corrective actions not feasible:
  - Notifying the Labor Inspection Office, then
  - RPE use allowed

<- Note: Involving one of the designated “experts” required.

↓

- Personal monitoring needed for selection of appropriate RPE
  - # of samples: 2+
  - Sampling duration: during the task
  - Result evaluation
    - Non-8hr-averaging
    - Comparing the highest value vs. “Control concentration”

No. 5: “Confirmation Measurement”

- For chemicals with Concentration Limit Values (CLVs)
- Measurement is NOT mandatory
  - Qualitative risk assessment is recommended first
  - Needed IF; Estimated exposure > CLV x 1/2
- # of samples
  - 1: the worst case
  - Or, 2+ : random selection from SEG
- Sampling duration: 8hr (minimum 2hrs)

- People almost always choose this, although the worst exposure is not easily identified.
- This represents to “compliance monitoring”

- STEL-measurement (15 min.)

No. 5: “Confirmation Measurement”

- Result evaluation
  - 8hr-averaging
  - Comparing “the largest value” vs. CLV
  - Acceptable/ Non-acceptable

- This discourages increasing # of samples.

- “Real” personal monitoring introduced, FINALLY
- The above method is only recommended: other method may be used.
- Concern: Employers tend to monitor minimum number (1 or 2) of samples.

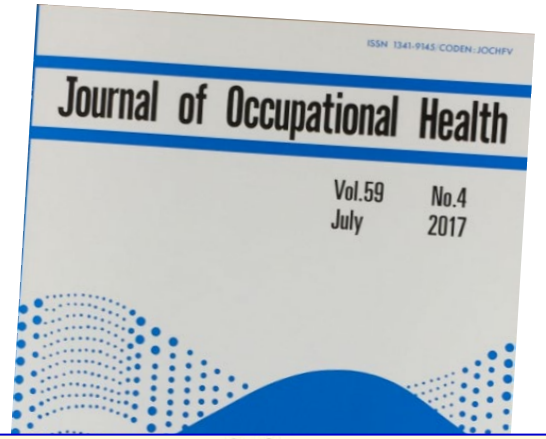
No. 6: General exposure measurement

- For chemicals subject to risk assessment requirements
  - Except those having CLVs
- Measurement is NOT mandatory
  - Depending on the qualitative risk assessment result
- Measurement method: at the employer's discretion.
  - In practice, “JSOH Method” commonly selected.

Japan Society for Occupational Health (JSOH)

No. 6: General exposure measurement - JSOH Method

- "Personal Exposure Monitoring Guideline for Chemicals"
  - Published in 2015
  - English version available
    Guidelines for personal exposure monitoring of chemicals: Part I | Journal of Occupational Health | Oxford Academic
  - Provided to Taiwan in 2018
  - Widely used, good reputation



J Occup Health 2017; 59: 367-373

Journal of Occupational Health

Special Article

Guidelines for personal exposure monitoring of chemicals: Part I

Expert Division of Occupational Hygiene & Ergonomics, the Japan Society for Occupational Health, "The Committee for Personal Exposure Monitoring"
Haruo Hashimoto[1], Kenichi Yamada[2], Hajime Hori[3], Shinji Kumagai[3], Masaru Murata[4], Toshio Nagoya[5], Hirohiko Nakahara[6] and Nobuyuki Mochida[7]

[1]University Management Center, Tokyo Institute of Technology, [2]Occupational Health Research and Development Center, Japan Industrial Safety and Health Association, [3]Department of Environmental Management, School of Health Sciences,

No. 6: General exposure measurement - JSOH Method

- Target Chemicals: open
- # of samples
  - Free: Selection from SEG (e.g. 2 to 6)
- Sampling time: 8hr (minimum, 2hrs)
- Result evaluation
  - 8hr-averaging
  - Classification based on Statistic analysis
    - Class 1(A, B, C), 2(A, B), 3

Non-restrictive personal monitoring

- Similar to AIHA, EU methods

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Future Challenges

1. Promote awareness among employers for “Autonomous management”
2. Widespread implementation of personal monitoring
   - ◆ Lowering measurement costs: e.g. 500 to 100 USD/sample, currently
     - Uncommon samples: unfamiliar with analysis
3. Support for SMEs
   - ◆ Establishing “paid consulting” system for Experts (COHs, Consultants)
     - Conventionally, “free consulting” common
4. Professional ethics regarding workplace measurement
   - ◆ Stop “cheating” on measurements
     - Training for professionals
5. Advancing technology nationwide, and improving quality of technicians (WEMs) /experts

Summary

A DRASTIC Regal Change in Chemical Substance Management in Japan
- Current Status -

Personal viewpoint

- Good points
  - Autonomous management started
  - Spreading understanding on “risk assessment”
  - Personal monitoring officially introduced
- Shortcomings, challenges
  - Double standards: conventional management still exists
  - Government regulating in excessive detail
  - Monitoring methods/cases too complex
  - SMEs: low awareness

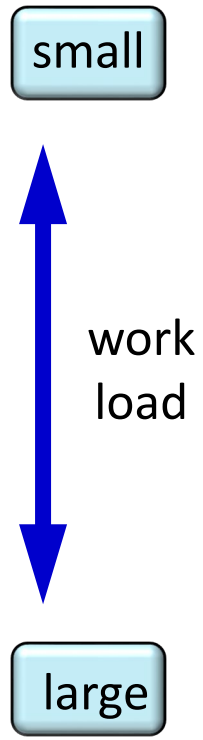


Back-up Slides

Risk assessment

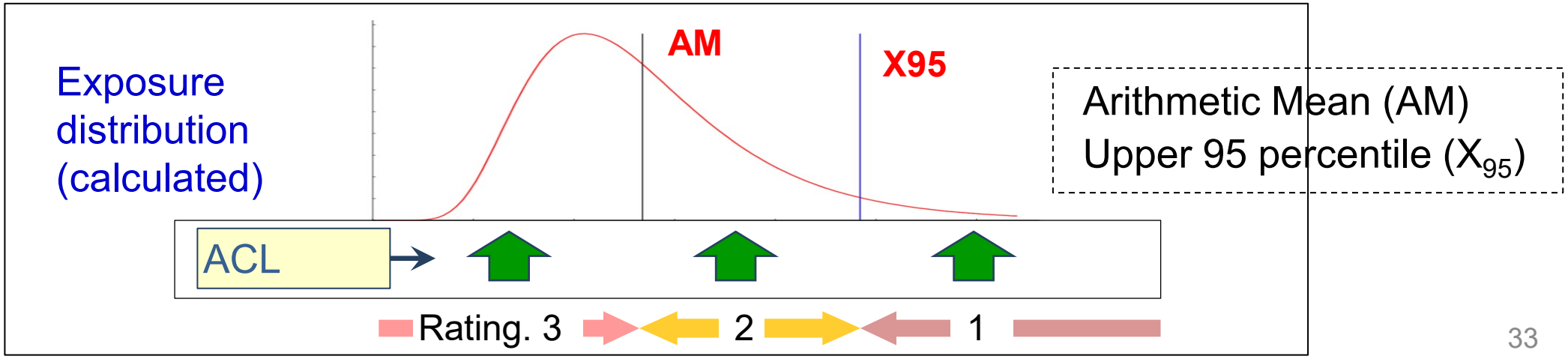
- There are several "methods" for this.
  - Methods for evaluating "exposure" directly or indirectly.

No.	method	Content
A	Decision based on	Past records, experience, technical common sense,
B	Simple risk assessment tool	CREATE- SIMPLE
C	Estimation using mathematical models	Calculated from a diffusion model of chemicals
D	Simple measurement	Detector tubes, direct-reading meters, etc.
E	Personal/area monitoring	WEM, Personal exposure monitoring



Result Evaluation: “A–Measurement”

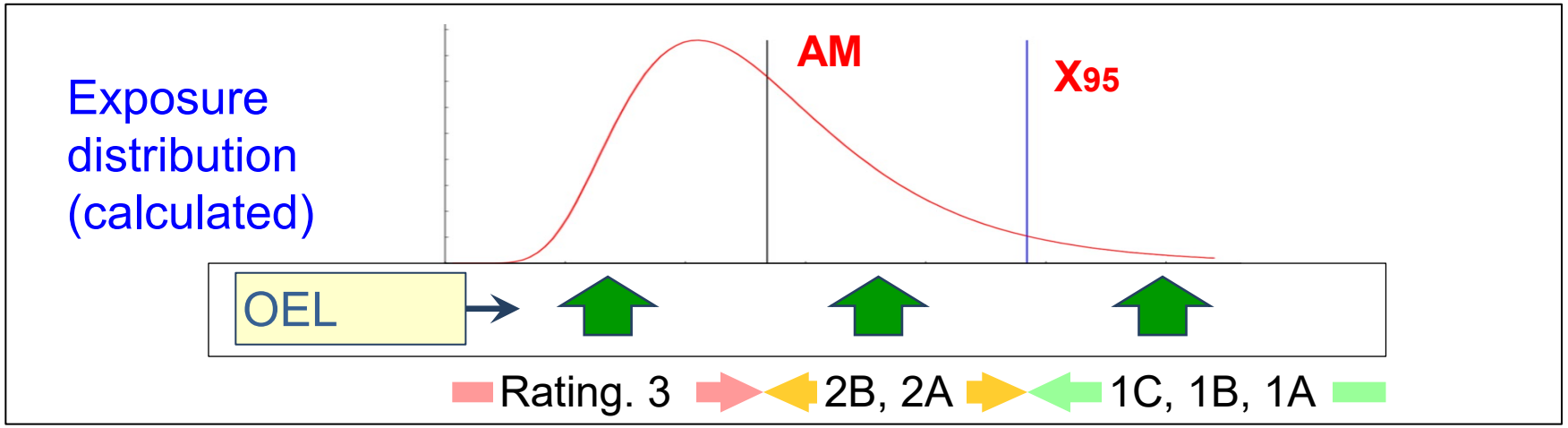
- Reference value: “Administrative Concentration Levels (ACL)”
  - Legally defined
  - Derived from ACGIH-TLV etc.
- Result rating
  - Three ratings: 1, 2, 3
  - Rating 3: ACL < AM



JSOH Method: Result Rating

- THREE-rating system, same as "A- measurement", basically
  - Subdivided into SIX ratings
- Welcomed by users

Rating	Definition		Indication
1A	X_{95} < OEL and	X_{95}< (OEL*10%)	Controlled (+)
1B		AM < (OEL*10%)	Controlled
1C		(OEL*10%) ≤ AM	Acceptable
2A	AM ≤ OEL ≤ X_{95} and	AM ≤ (OEL*50%)	Exposure reduction recommended
2B		(OEL*50%) < AM	Uncontrolled
3	OEL < AM		Uncontrolled (-)



【STEP 3】諸条件の入力

◆ Q&A方式（Q1 ~ Q7）

【STEP 3】以下の作業内容に関する質問に答えましょう。

Q1　製品の取扱量はどのくらいですか。

中量　（1L以上～1000L未満）

Q2　スプレー作業など空気中に飛散しやすい作業を行っていますか。

いいえ

Q3　化学物質を塗布する合計面積は1m2以上ですか。

いいえ

Q4　作業場の換気状況はどのくらいですか。

換気レベルC（工業的な全体換気、屋外作業）

Q5　1日あたりの化学物質の作業時間（ばく露時間）はどのくらいですか。

30分以下

Q6　化学物質の取り扱い頻度はどのくらいですか。

週1回以上　➡　5　日／週

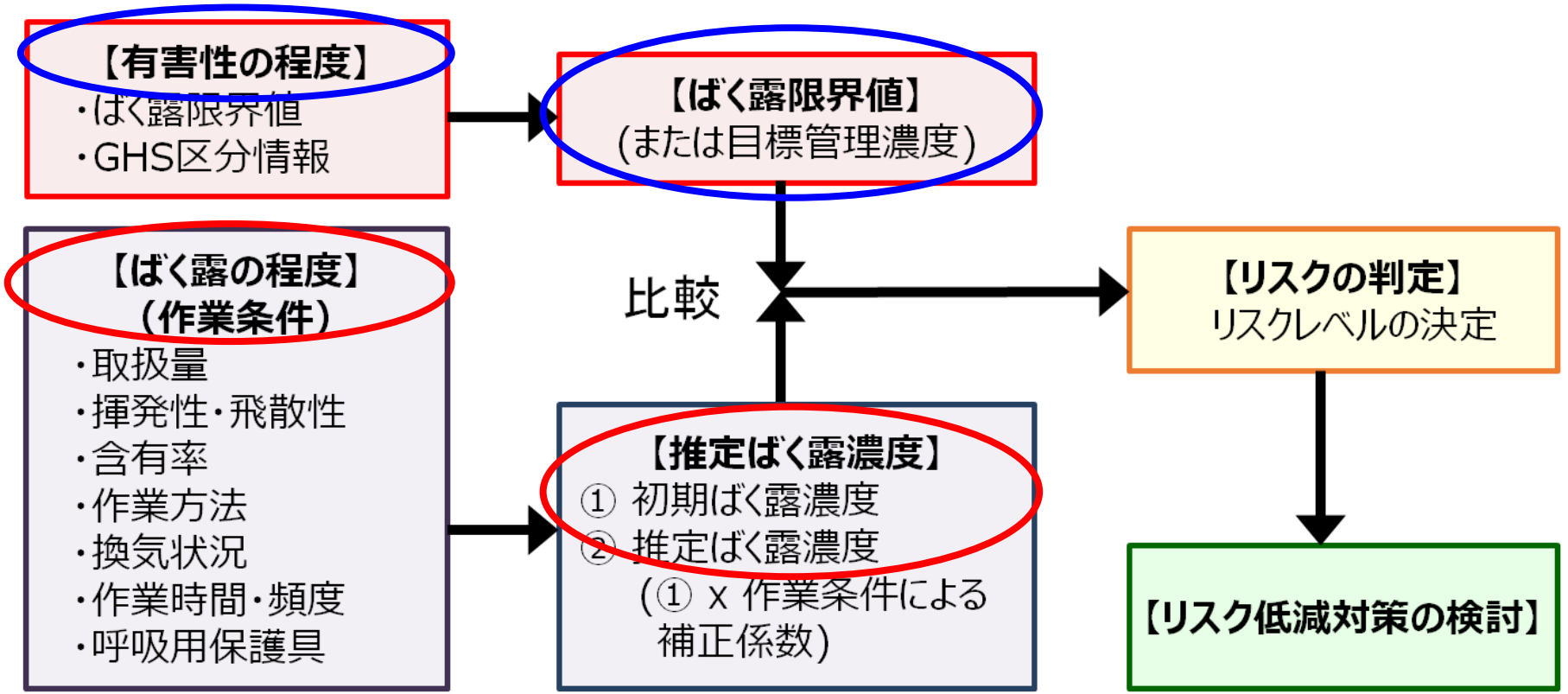
Q7　作業内容のばく露濃度の変動の大きさはどのくらいですか。

ばく露濃度の変動が小さい作業

手入力

CREATE-SIMPLEの仕組み

◆「ばく露評価（ばく露濃度の推定）」を行い，「ばく露限界値」と比較するもの



判断要素の数が「8」と多い → 推定の精度が比較的高い

【STEP 3】諸条件の入力

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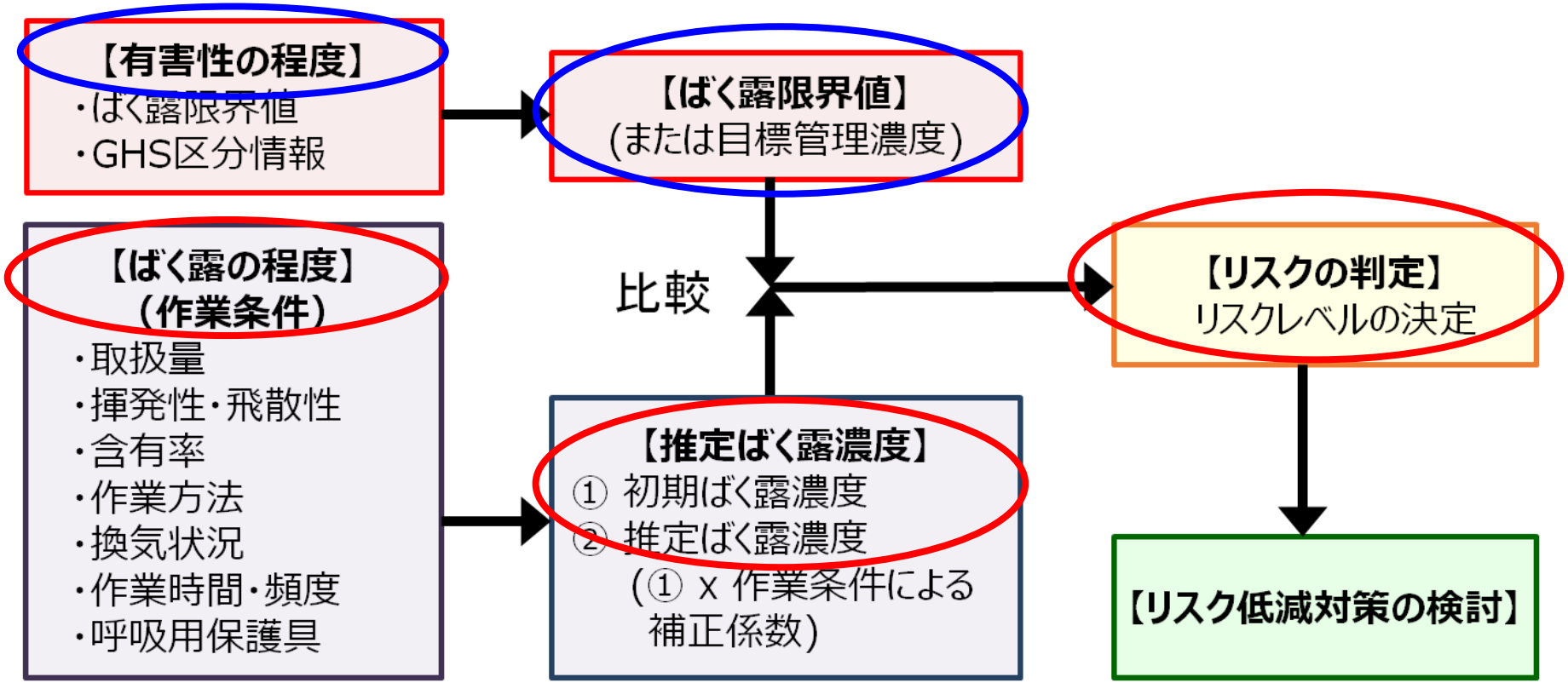
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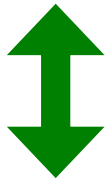
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【STEP 4】リスクの判定ーその仕組み

- CREATE-SIMPLEは「ばく露濃度を推定」
  - 1桁の幅の濃度として提示（8時間ばく露）
  - この「上側値」を，



	推定ばく露濃度	
	吸入（8時間）	吸入（短時間）
	5～50 ppm	2000 ppm

← この場合は，
- 上側値 = 50 ppm
- ばく露限界値 = 20 ppm



リスクレベル = Ⅲ

- ばく露限界値と比較

	ばく露限界値（管理目標濃度）	
	吸入（8時間）	吸入（短時間）
108-88-3	トルエン	
	20 ppm	100 ppm

リスクレベル	定義（OEL=ばく露限界値）
Ⅳ（大きなリスク）	10 x [OEL] < 上側値
Ⅲ（中程度のリスク）	[OEL] < 上側値
Ⅱ-B（小さなリスク）	0.5 x [OEL] < 上側値 ≦ [OEL]
Ⅱ-A（小さなリスク）	上側値 ≦ 0.5 x [OEL]
Ⅰ（些細なリスク）	上側値 ≦ 0.1 x [OEL]